Summary

- Linear programming problem(LPP) is a mathematical modeling technique which is used to allocate limited available resources in order to optimize (maximize or minimize) the objective function.
- **Short form of LPP**

  Maximize or Minimize $Z = \sum_{j=1}^{n} c_j x_j$

  Subject to $\sum_{j=1}^{n} a_{ij} x_j \leq (\text{or} = \text{or} \geq) b_i,\ i = 1, 2, 3, \ldots, m$

  and $x_j \geq 0$
- **Objective function:** A function $Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n$ which is to be optimized (maximized or minimized) is called objective function.
- **Decision variable:** The decision variables are the variables we seek to determine to optimize the objective function. x_j , $j = 1,2,3,\ldots,n$, are the decision variables.
- **Solution:** A set of values of decision variables $x_j, j = 1,2,3,\ldots,n$ satisfying all the constraints of the problem is called a solution to that problem.
- **Feasible solution:** A set of values of the decision variables that satisfies all the constraints of the problem and non-negativity restrictions is called a feasible solution of the problem.
- **Optimal solution:** Any feasible solution which maximizes or minimizes the objective function is called an optimal solution.
- **Feasible region:** The common region determined by all the constraints including non-negative constraints $x_j \geq 0$ of a linear programming problem is called the feasible region (or solution region) for the problem.
- Linear programming problems which involve only two variables can be solved by graphical method.
- It should be noted that the optimal value of LPP occurs at the corner points of the feasible region
- Network is a diagrammatic representation of various activities concerning a project arranged in a logical manner.

10_11th_BM-STAT_Ch-10-EM.indd 253 21-04-2020 12:33:02 PM